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Vehicle

Abstract

A vehicle includes: a vehicle body including an opening part located in a rear part of the vehicle; a hatch supported by the vehicle body so as to be able to open and close the opening part; an under hatch component located inside the vehicle with respect to the opening part; and a rack movably supported by the vehicle body between a first position facing the under-hatch component and a second position retracted from the first position. The under-hatch component is exposed outside through the opening part in a state where the hatch is opened and the rack is located in the second position.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4573854	12/1985	McFarland	267/179	A61G 3/0209
7918465	12/2010	Metzger	280/439	B60P 1/027
8322772	12/2011	Gilbeck et al.	N/A	N/A
2020/0148047	12/2019	Kay	N/A	B60R 5/04

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
115139721	12/2021	CN	B60G 17/0152
102008056375	12/2020	DE	B60J 7/146
1787613	12/2006	EP	A61G 3/061
2911361	12/2007	FR	E05F 15/63
100580004	12/2005	KR	B62D 25/087
WO-8907569	12/1988	WO	N/A

OTHER PUBLICATIONS

Chen, CN-115139721-A, Machine Translation of Specification (Year: 2022). cited by examiner
Dakkuya, FR-2911361-A1, Machine Translation of Specification (Year: 2008). cited by examiner
Bunsmann, DE-102008056375-B4, Machine Translation of Specification (Year: 2021). cited by examiner

Jeongho, KR-100580004-B1, Machine Translation of Specification (Year: 2006). cited by examiner
Pawl, WO-8097569-A1, Machine Translation of Specification (Year: 1989). cited by examiner

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a vehicle with a rack.

BACKGROUND ART

(2) U.S. Pat. No. 8,322,772 BI discloses a vehicle with a cargo bed.

SUMMARY

(3) The present vehicle is a vehicle including: a vehicle body including an opening part located in a rear part of the vehicle; a hatch supported by the vehicle body so as to be able to open and close the opening part; an under hatch component located inside the vehicle with respect to the opening part; and a rack movably supported by the vehicle body between a first position facing the under-hatch component and a second position retracted from the first position, wherein the under-hatch component is exposed outside through the opening part in a state where the hatch is opened and the rack is located in the second position.

(4) According to the present vehicle, the rack is covered by the hatch, thus a stored object stored in the rack can be protected from rain water, for example. In a state where the hatch is closed and the rack is located in the first position, the under-hatch component can be protected by the rack and the hatch, and a stored object loaded on the rack can be protected by the hatch. Furthermore, in the state where the hatch is opened and the rack is located in the second position, it is easy to have access to the under-hatch component through the opening part.

(5) The objects, features, aspects, and advantages of this disclosure will become more apparent from the following detailed description and the accompanying drawings.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a side view illustrating a vehicle.
- (2) FIG. 2 is a side view illustrating an inner part of a hardtop in the vehicle.
- (3) FIG. 3 is a side view illustrating a state where a hatch is developed.
- (4) FIG. 4 is a side view illustrating a state where the hatch and a rack are developed.
- (5) FIG. 5 is a perspective view illustrating a rear part of the vehicle.
- (6) FIG. 6 is a perspective view illustrating the state where the hatch is developed.
- (7) FIG. 7 is a perspective view illustrating the state where the hatch and the rack are developed.
- (8) FIG. 8 is a block diagram illustrating an apparatus configuration of the vehicle.
- (9) FIG. 9 is a flow chart illustrating an example of a control operation of an ECU.
- (10) FIG. 10 is a side view illustrating a vehicle in which a vehicle body height is adjusted.
- (11) FIG. 11 is a flow chart illustrating another example of the control operation of the ECU.
- (12) FIG. 12 is a flow chart illustrating still another example of the control operation of the ECU.
- (13) FIG. 13 is a block diagram illustrating an apparatus configuration of the vehicle.
- (14) FIG. 14 is a plan view illustrating a lock/release state of a rack lock.
- (15) FIG. 15 is a flow chart illustrating an example of a control operation of an ECU.
- (16) FIG. 16 is a side view illustrating a modification example of a position of an operating element.
- (17) FIG. 17 is a side view illustrating a modification example of a vehicle body.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

(18) A vehicle according to an embodiment will be hereinafter described. FIG. 1 is a side view illustrating a vehicle 10. FIG. 2 is a side view illustrating an inner part of a hardtop 16 in the vehicle 10. In the following description, a travel direction of the vehicle 10 may be expressed as FRONT, a direction opposite to the travel direction may be expressed as REAR, a road surface side on which the vehicle 10 travels may be expressed as LOW, and a side opposite to the road surface side may be expressed as UP. The RIGHT-LEFT may be referred to while a person stands on a road surface and faces forward. A right-left direction is a vehicle-width direction.

(19) <Whole Configuration of Vehicle>

(20) The vehicle 10 includes a vehicle body 11, a hatch 40, an under-hatch component 32, and a rack 50. In the present embodiment, the vehicle 10 further includes a seat 26, a travel mechanism,

and a drive operation mechanism.

(21) The vehicle body **11** includes an opening part **19** (refer to FIG. **6**) located in a rear part of the vehicle **10**. In the present embodiment, the vehicle body **11** includes a rollover protective structure (ROPS) **12**, the hardtop **16**, and a chassis **20**. In the present embodiment, the opening part **19** is formed on an upper side of the rear part of the hardtop **16**.

(22) The ROPS **12** determines an occupant space. The ROPS **12** is formed by connecting a plurality of frame parts. A part of the plurality of frame parts located in front of the vehicle **10** determines an occupant space. A part of the plurality of frame parts located in back of the vehicle **10** determines a luggage space and a space for the under-hatch component **32**. The plurality of frame parts include a lateral frame part **13** mainly extending in a front-back direction, a vertical frame part **14** mainly extending in a vertical direction, and a cross frame part **15** mainly extending in a right-left direction. Each of the lateral frame parts **13** and the vertical frame parts **14** is basically provided on right and left sides in a pair. The cross frame part **15** connects the pair of lateral frame parts **13** or the pair of vertical frame parts **14** to each other. In the right-left direction, the seat **26** and a steering wheel **34**, for example, are located between the pair of lateral frame parts **13** and the pair of vertical frame parts **14**.

(23) In the example illustrated in FIG. **2**, the lateral frame part **13** includes a lateral frame part **13A** extending in the front-back direction on an upper side of the seat and a lateral frame part **13B** extending in the front-back direction in back of the seat, for example. The vertical frame part **14** includes a vertical frame part **14A** extending in an up-down direction in back of the seat and a vertical frame part **14B** extending in the up-down direction in back of the vehicle **10**, for example. The cross frame part **15** includes a cross frame part **15A** in the right-left direction on the upper side of the seat and a cross frame part **15B** extending in the right-left direction in back of the vehicle **10**, for example.

(24) A region between the plurality of frame parts in the ROPS **12** is opened. An opening between the plurality of frame parts in the ROPS **12** in the vehicle **10** may be appropriately covered by a cover, for example.

(25) A member covering an opening in front of the seat **26** in the cover provided to the ROPS **12** can be considered a windshield. A member covering an opening on the upper side of the seat **26** in the cover provided to the ROPS **12** can be considered a roof. A member covering an opening on a lateral side of the seat **26** in the cover provided to the ROPS **12** can be considered a door **18C**. A member covering an opening in back of the seat **26** in the cover provided to the ROPS **12** can be considered a partition wall **22**. One or some of the windshield, the roof, the door **18C**, and the partition wall **22** may be provided as the cover provided to the ROPS **12**. When some of the windshield, the roof, the door **18C**, and the partition wall **22** are provided as the cover, plural parts may be individually or integrally provided. Provided in the present embodiment is the hardtop **16** integrally including the windshield, the roof, the door **18C**, and the partition wall **22** as the cover. In the present embodiment, a major part of the ROPS **12** is covered by the hardtop **16**.

(26) The hardtop **16** is made of metal or rigid resin, and has high rigidity. In the example in FIG. **1**, the hardtop **16** includes a roof part **17** and a sidewall part **18**. The roof part **17** is located on the upper side of the vehicle **10**. The roof part **17** covers the upper side of the seat **26**. The sidewall part **18** is located on both lateral sides of the vehicle **10**. The sidewall part **18** includes a front sidewall part **18A** and a rear sidewall part **18D**.

(27) The front sidewall part **18A** is located on a lateral side of the seat. A ride opening part **18B** is formed in the front sidewall part **18A**. An occupant can get in and out of the vehicle **10** through the ride opening part **18B**. Herein, the door **18C** is attached to the ride opening part **18B**. The door **18C** can open and close the ride opening part **18B**. The vehicle **10** may be configured to achieve normal traveling while the ride opening part **18B** is not completely covered by the door **18C** but remains open.

(28) The rear sidewall part **18D** extends to a rear side from the front sidewall part **18A**. The rear

sidewall part **18D** protrudes from a part of the front sidewall part **18A** along a height direction. A portion surrounded by upper edges of the pair of rear sidewall part **18D** and a rear edge of the roof part **17** is the opening part **19** opened and closed by the hatch **40**.

(29) In the vehicle **10**, each part (the seat **26**, the ROPS **12**, a travel drive source, **32**, wheels **30F** and **30R**, for example) is supported by the chassis **20**.

(30) The hatch **40** is supported by the vehicle body **11** so as to be able to open and close the opening part **19**. FIG. **1** illustrates a state where the hatch **40** is closed. The under-hatch component **32** is located inside the vehicle **10** with respect to the opening part **19**. The rack **50** is movably supported by the vehicle body **11** between a first position and a second position. The first position is a position where the rack **50** faces the under-hatch component **32**. The second position is a position retracted from the first position. In a state where the hatch **40** is opened and the rack **50** is located in the second position, the under-hatch component **32** is exposed outside through the opening part **19**.

(31) The seat **26** includes a seat cushion **27** in which an occupant is seated and a seat back **28** on which the occupant leans. The seat back **28** has a height high enough to support a back of the occupant seated in the seat **26**. In the example illustrated in FIG. **2**, the seat **26** is provided in only one row in the front-back direction. The seat **26** may be separately provided in two rows in the front-back direction, or may also be provided in three or more rows.

(32) The travel mechanism is a mechanism for the vehicle **10** to travel, and includes the wheels **30F** and **30R**, a travel drive source **32**, and a power transmission mechanism. Rotation drive force of the travel drive source **32** is transmitted to the wheels **30F** and **30R** through the power transmission mechanism, thus the vehicle **10** travels. Described in the present embodiment is an example that the vehicle **10** includes four wheels **30F** and **30R**. The vehicle **10** may include three wheels, or may also include six wheels. The configuration of the power transmission mechanism is discretionary. Examples of the power transmission mechanism may include a transmission and a drive shaft.

(33) The wheels **30F** and **30R** include front wheels **30F** and rear wheels **30R**. Two front wheels **30F** are rotatable around its own central axis and are supported to be rotatable around a steering axis at right and left front portions of the chassis **20**. Two rear wheels **30R** are supported to be rotatable around its own central axis at right and left rear portions of the chassis **20**.

(34) The travel drive source **32** may be an internal combustion engine that burns fuel in a combustion chamber and generates power for traveling through expansion of gases caused by burning. In this case, it is sufficient that the vehicle **10** includes a fuel tank. The travel drive source **32** may be an electric motor that converts electric energy into rotational motion for traveling. In this case, it is sufficient that the vehicle **10** includes a traveling battery for supplying the electric energy to the travel drive source **32**. A position of the travel drive source **32** is discretionary. Herein, the travel drive source **32** is supported in the rear part of the vehicle **10**.

(35) The drive operation mechanism is a mechanism for receiving a drive operation of a driver, and includes a steering wheel **34** and a pedal not shown in the diagrams. The driver operates the steering wheel **34**, thereby being able to rotate the front wheels **30F** around the steering axis. The driver operates the pedal, thereby being able to perform an acceleration operation and a brake operation. The steering wheel **34** and the pedal are located in front of the seat **26**. The driver seated in the seat **26** can operate the steering wheel **34** and the pedal. It is applicable that the pedals are an accelerator pedal and a brake pedal that are separately provided, or both the acceleration and the brake can be operated with one pedal.

(36) The wheels **30F** and **30R** are supported by the vehicle body **11** via a suspension **36**. The suspension **36** generally includes a suspension arm **37**, a spring, and a shock absorber **38**. The spring and the shock absorber **38** may be integrally formed. A component in which the spring and the shock absorber **38** are integrally formed is also referred to as a coil over, for example. The suspension **36** may be an axle suspension system (rigid type) in which the wheels on the right and left sides are integrally moved, or may also be an independent suspension system (independent

type) in which the wheels on the right and left sides are independently moved. When the vehicle **10** is an off-road vehicle, the suspension **36** is preferably the independent suspension system.

(37) The independent suspension system may be any of various types of suspension systems such as a double wishbone system, a strut system, a multilink system, or a trailing-arm system. The double wishbone system includes an upper arm and a lower arm pivotable around a wheel hub as the suspension arm **37**. The strut system includes a lower arm as the suspension arm **37** and does not include an upper arm. The multilink system includes one or more arms in addition to an upper arm and a lower arm as the suspension arm **37**. The trailing-arm system includes a swing arm including a swing axis (pivot). The swing arm is rigidly connected to a wheel hub.

(38) The same suspension system may be adopted to the suspension **36** of the front wheels **30F** and the suspension **36** of the rear wheels **30R**. Different suspension systems may be adopted to the suspension **36** of the front wheels **30F** and the suspension **36** of the rear wheels **30R**. For example, it is applicable that the suspension **36** of the front wheels **30F** is the double wishbone system and the suspension **36** of the rear wheels **30R** is the trailing-arm system.

(39) The suspension **36** may be an active suspension or a passive suspension. The active suspension is a suspension which can control damping force by hydraulic pressure or pneumatic pressure, for example. In this case, the vehicle **10** includes a control apparatus electrically controlling the active suspension. The active suspension may be a hydraulic pressure active suspension provided with a hydraulic pressure actuator in which hydraulic pressure can be adjusted as a shock absorber, or may also be an air suspension provided with an air chamber in which pneumatic pressure can be adjusted as a spring, for example. The passive suspension is a suspension which does not include a mechanism which can control damping force by hydraulic pressure or pneumatic pressure, for example.

(40) A control system of the active suspension may be semi-active control or full-active control. In the semi-active control, only damping force while the vehicle is traveling is controlled. In the full-active control, a vehicle body height while the vehicle is stopping is controlled in addition to the damping force while the vehicle is traveling. The active suspension on which the full-active control can be performed functions as a vehicle body height adjustment apparatus. A control apparatus in which the full-active control is performed on the active suspension functions as a control apparatus controlling the vehicle body height adjustment apparatus. In the present embodiment, the suspension is the active suspension on which the full-active control can be performed.

(41) The vehicle **10** includes an electronic control unit (ECU) **39** as the control apparatus, for example. The ECU **39** controls each apparatus in the vehicle **10**. The number of the ECUs **39** may be one, or the plurality of ECUs **39** are also applicable. Control target apparatuses controlled by the plurality of ECUs **39** may be divided in accordance with types of apparatuses, or may also be divided in accordance with an area where the apparatuses are disposed. An arrangement position of the ECU **39** is particularly limited but can be appropriately set. Herein, the ECU **39** is disposed in a space in front of a driver seat. For example, the ECU **39** may be disposed in a space in an instrument panel, or may also be disposed in a luggage space.

(42) <Hatch and Rack>

(43) FIG. **3** is a side view illustrating a state where the hatch **40** is developed. FIG. **4** is a side view illustrating a state where the hatch **40** and the rack **50** are developed. FIG. **5** is a perspective view illustrating a rear part of the vehicle **10**. FIG. **6** is a perspective view illustrating the state where the hatch **40** is developed. FIG. **7** is a perspective view illustrating a state where the hatch **40** and the rack **50** are developed.

(44) The hatch **40** includes a hatch body **41**, a hatch hinge **42**, and a hatch handle **43**. Herein, the hatch **40** is supported by the ROPS **12**.

(45) The hatch body **41** includes a rear wall part **41A** extending along the up-down direction and the right-left direction. Herein, the hatch body **41** includes an upper wall part **41B** extending from an upper end of the rear wall part **41A** toward the front-back direction. The rear wall part **41A** of

the hatch body **41** has a posture along the up-down direction in a close state, and changes the posture toward a posture along a horizontal direction when the hatch body **41** is opened.

(46) The hatch hinge **42** is located in an end portion of the hatch body **41**. The hatch hinge **42** is connected to the vehicle body **11**. Herein, the hatch hinge **42** is located in a front end of the upper wall part **41B**. The hatch hinge **42** is connected to the cross frame part **15A** of the ROPS **12**. The hatch hinge **42** connects the hatch body **41** to the vehicle body **11** to be rotatable around a rotation axis of the hatch **40**. The rotation axis of the hatch **40** is an axis extending along the right-left direction.

(47) The hatch handle **43** is located in a rear part of the hatch body **41**. The hatch handle **43** is provided on a rear surface of the rear wall part **41A**.

(48) The hatch **40** may be openable and closable by a hatch latch and a hatch striker to which the hatch latch is locked. When the hatch latch is locked to the hatch striker, the hatch body **41** is kept in a close state. One of the hatch latch and the hatch striker is provided to the hatch **40**, and the other one thereof is provided to a member which is integrally formed with the hatch **40** and is not moved. The member which is integrally formed with the hatch **40** and is not moved may be the vehicle body **11** or the rack **50**. For example, it is applicable that the hatch latch is provided to a peripheral edge of the rear wall part **41A**, and the hatch striker is provided to the vehicle body **11**. The hatch striker may be located below a lower edge of the rear wall part **41A**, for example. The hatch striker may be provided to the cross frame part **15B** or a member (rack body **51**, for example) supported by the cross frame part **15B**, for example.

(49) The hatch latch is opened and closed by an open-close operating element. When the open-close operating element pivots the hatch latch, a lock state on the hatch striker is released. The open-close operating element includes a latch lever connected to a shaft of the latch and an external operation mechanism operating the latch lever by an operation from outside, for example. It is applicable that the hatch latch can be operated only manually, or can be operated electrically or manually.

(50) For example, the hatch handle **43** may have a function as an external operation mechanism. For example, it is also applicable that when the hatch handle **43** is pivotably supported by the rear wall part **41A** and is pivoted, the latch lever connected to the hatch handle **43** pivots the latch, thereby being able to release the lock on the striker. The external operation mechanism may be provided separately from the hatch handle **43**. For example, the external operation mechanism may include a switch such as a push button, a wiring connected to the switch, and a motor connected to the wiring to pivot the latch lever. In this case, the switch may be provided to the hatch handle **43**. The switch may be provided in a position away from the hatch handle **43** such as a driver seat.

(51) A hatch lock may be provided to the hatch **40**. The hatch lock locks the hatch **40** in a close state. The hatch lock may lock the hatch latch described above. In this case, for example, the hatch lock includes a locking lever locked to the latch lever. The hatch lock may be provided separately from the hatch latch. The hatch lock includes an external operation mechanism performing a lock/release operation by an operation from outside. The external operation mechanism may include a switch such as a push button, a wiring connected to the switch, and a motor connected to the wiring to pivot the lock. In this case, the switch may be provided to the hatch handle **43**. The switch may be provided in a position away from the hatch handle **43** such as a driver seat. It is applicable that the lock/release operation of the hatch lock can be operated only manually, or can be operated electrically or manually.

(52) For example, a component such as a door lock apparatus in which the latch, the latch lever, and the locking lever are integrally formed can be used as a component in which the hatch latch, the open-close operating element of the hatch latch, and the hatch lock are integrally formed.

(53) As illustrated in FIG. **6**, when the hatch **40** is opened, the opening part **19** is released. The hatch **40** is moved to an upper position beyond a pivot range of the rack **50** while the opening part **19** is released. The hatch **40** in the close state interferes with the pivot range of the rack **50**.

(54) The vehicle **10** includes a hatch holder holding the hatch **40** in a position where the hatch **40** is opened. The hatch holder also functions as a bias member biasing the hatch **40** to an opening direction. The vehicle **10** includes a hatch dumper **46** as the hatch holder. The hatch dumper **46** can also suppress abrupt opening and closing of the hatch **40**. The hatch dumper **46** includes a cylinder housing a fluid, a piston moved in the cylinder, and a cylinder rod extending from the piston toward outside the cylinder. One of the cylinder and the cylinder rod is fixed to the hatch **40**, and the other one thereof is fixed to the vehicle body **11**, for example. Then, when the hatch **40** is opened and closed, a position of the piston is changed, and pressure in the cylinder is changed. For example, when the fluid is in a compressed state while the hatch **40** is in the close state, the hatch **40** is biased to the opening direction. When pressure of the fluid is applied in the movement of the piston, abrupt movement of the hatch **40** is suppressed. The hatch dumper **46** may not include a mechanism of changing the pressure in the cylinder other than the open-close operation of the hatch **40**.

(55) A luggage space is formed in a lower side of the hatch **40**. An occupant space is formed in a front side of the luggage space. The luggage space is a space in which the rack **50** is provided. The occupant space is a space in which the seat is provided. Herein, the vehicle body **11** includes the partition wall **22** partitioning the luggage space and the occupant space. The partition wall **22** extends along the up-down direction and the right-left direction in the rear part of the seat **26**. The partition wall **22** may not be provided.

(56) The rack **50** includes the rack body **51** and a rack hinge **53**.

(57) The rack body **51** includes a mounting surface **51A** to which a stored object **60** is mounted. The mounting surface **51A** of the rack body **51** has a posture along the horizontal direction in the first position, and has a posture along the vertical direction when the mounting surface **51A** is moved toward the second position. A position illustrated in FIG. **4** is the second position herein.

(58) The rack hinge **53** is located in an end portion of the rack body **51**. The rack hinge **53** is connected to the vehicle body **11**. Herein, the rack hinge **53** is located in a rear end of the rack body **51**. The rack hinge **53** is connected to the cross frame part **15B** of the ROPS **12**. The rack hinge **53** connects the rack body **51** to the vehicle body **11** to be rotatable around a rotation axis of the rack **50**. The rotation axis of the rack **50** is an axis extending along the right-left direction.

(59) The rack **50** may have a horizontal posture of being further rotated at substantially 90 degrees from the position illustrated in FIG. **4** so that a surface thereof on a side opposite to the mounting surface **51A** is directed to the upper side, for example. The rack **50** may have a vertical posture of being further rotated at substantially 180 degrees from the position illustrated in FIG. **4** so that the mounting surface **51A** is directed to the front side. The rack **50** may be configured to be able to be positioned at a third position between the first position and the second position.

(60) Herein, the rack **50** is a tire rack **50** to which a spare tire **60** is mounted. The tire rack **50** can be moved between the first position and the second position while the spare tire **60** is mounted thereto.

(61) The rack **50** may include a fixing component **52** attached to the rack body **51** to fix the stored object **60**. Herein, a fixing belt is provided to the rack **50** as the fixing component **52** to fix the spare tire **60**.

(62) The rack **50** may be the rack **50** other than the tire rack **50**. For example, a cool box or a luggage loading box may be mounted to the rack **50**.

(63) A range **R1** illustrated in FIG. **4** is a pivot range of the spare tire **60** mounted to the rack **50** between the first position and the second position. Herein, the hatch **40** is moved to the upper side beyond the range **R1** while the opening part **19** is released. The range **R1** interferes with the hatch **40** in the close state.

(64) The vehicle **10** may include a movement mechanism **54** of moving the rack **50** between the first position and the second position. The movement mechanism **54** may be a cylinder device, for example. The cylinder device includes a cylinder housing a fluid, a piston moved in the cylinder, a

cylinder rod extending from the piston toward outside the cylinder, and a mechanism changing pressure in the cylinder, for example. One of the cylinder and the cylinder rod is fixed to the rack **50**, and the other one thereof is fixed to the vehicle body **11**, for example. Then, when pressure in the cylinder is changed, a position of the piston is changed, and the rack **50** is moved. The cylinder device also functions as a rack holder holding the rack **50** in a position where the rack **50** is opened. (65) The vehicle **10** may not include the movement mechanism **54** of moving the rack **50** between the first position and the second position. In this case, the rack **50** is moved by manpower. In this case, it is sufficient that a rack handle for a person to grip the rack **50** is provided to the rack body **51**. In this case, it is sufficient that a rack dumper is provided. The rack dumper includes a cylinder, a piston, and a cylinder rod in the manner similar to the hatch dumper **46** described above, and does not include a mechanism of changing pressure in the cylinder. The rack dumper functions as a rack holder holding the rack **50** in the second position. The rack dumper also functions as a bias member biasing the rack **50** to the second position. The rack dumper can suppress abrupt movement of the rack **50**.

(66) The vehicle **10** includes an operating element **55** performing an open-close operation on the rack **50**. The operating element **55** is located inside the vehicle **10** with respect to the hatch **40** in the close state. As illustrated in FIG. **6**, the operating element **55** is provided on an inner surface of the rear sidewall part **18D**. The operating element **55** may be provided to an inner surface of the hatch body **41** or a rear surface of the partition wall **22**. For example, when the operating element **55** is operated, the mechanism of changing the pressure in the cylinder in the cylinder device described above is actuated, and the rack **50** is opened and closed.

(67) The under-hatch component **32** is located below the rack **50**. The under-hatch component **32** may include a component relating to a travel mechanism such as the travel drive source **32**, an oil filter, a battery, or an air cleaner. Herein, the travel drive source **32** constitutes the under-hatch component **32**. The travel drive source **32** may be an engine (internal combustion engine) or an electrical motor.

(68) The under-hatch component **32** may include a maintenance target component, for example. The maintenance target component is a component which requires regular maintenance or a component which has a high maintenance frequency, for example. For example, the component relating to the travel mechanism such as the travel drive source **32**, the oil filter, the battery, and the air cleaner described above is considered the maintenance target component. It is easy to have access to the maintenance target component, thus maintenance properties of the maintenance target component are improved. It is sufficient that a maintenance target part of the maintenance target component is provided in an accessible position through the opening part **19**.

(69) When the under-hatch component **32** includes the maintenance target component, it is applicable that the rack **50** can retract to a position which does not interfere with a maintenance operation of the maintenance target component, or can be detached from the vehicle body **11**.

(70) The under-hatch component **32** may include a plurality of components. A component having high design properties in the plurality of components may be provided in a position exposed more easily than a component having low design properties. A component having high maintenance properties in the plurality of components may be provided in a position exposed more easily than a component having low maintenance properties.

(71) The under-hatch component **32** may be exposed from a position other than the opening part **19**. For example, the under-hatch component **32** may be exposed from a position between the lower part of the rear sidewall part **18D** and the chassis **20**.

(72) <Control of Vehicle Body Height>

(73) FIG. **8** is a block diagram illustrating an apparatus configuration of the vehicle **10**.

(74) The vehicle **10** includes a hatch sensor **48**, a rack sensor **58**, a vehicle body height adjustment apparatus **36**, and a control apparatus **39**. The hatch sensor **48** detects rotation of the hatch **40**. The rack sensor **58** may detect rotation of the rack **50**. The rack sensor **58** may detect inclination of the

rack 50 with respect to the vehicle body 11 or horizon. The rack sensor 58 may include a sensor detecting the rotation of the rack 50 and a sensor detecting the inclination of the rack 50 with respect to the vehicle body 11 or horizon. The vehicle body height adjustment apparatus 36 adjusts a height of the vehicle body 11. The control apparatus 39 controls the vehicle body height adjustment apparatus 36. Herein, the active suspension 36 described above functions as the vehicle body height adjustment apparatus 36. The vehicle body height adjustment apparatus 36 may be made as a mechanism different from the suspension 36. Herein, the ECU 39 functions as the control apparatus 39 controlling the vehicle body height adjustment apparatus 36.

(75) A detection result of the hatch sensor 48 and a detection result of the rack sensor 58 are transmitted to the ECU 39. The ECU 39 controls the vehicle body height adjustment apparatus 36 based on the detection result of the hatch sensor 48 and/or the detection result of the rack sensor 58. The vehicle body height adjustment apparatus 36 is a hydraulic type apparatus or a pneumatic type apparatus, and the vehicle body 11 can be moved up and down by adjusting hydraulic pressure or pneumatic pressure. Herein, the active suspension 36 provided to each of four wheels functions as the vehicle body height adjustment apparatus 36. Accordingly, the vehicle body 11 can be moved up and down independently in a position corresponding to each of four wheels regardless of the other position.

(76) The vehicle body height adjustment may be inclination of the vehicle body 11 or moving up and down of the vehicle body 11 in a horizontal posture without inclination thereof. When the vehicle body 11 is inclined, the vehicle body 11 may be inclined so that heights of the front part and the rear part are different from each other. In this case, when the rear part is lower in height than the front part, the stored object 60 can be taken in and out from the rack 50. When the vehicle body 11 is inclined, the vehicle body 11 may be inclined in a rolling direction. That is to say, the vehicle body 11 may be inclined so that heights of a right part and a left part of the vehicle 10 are different from each other. In this case, the under-hatch component 32 can be easily seen.

(77) FIG. 9 is a flow chart illustrating an example of a control operation of the ECU 39. In the example illustrated in FIG. 9, the ECU 39 controls the vehicle body height adjustment apparatus 36 based on both a state of the hatch 40 and a state of the rack 50.

(78) In Step S11, the ECU 39 determines whether the open-close state of the hatch 40 is changed based on the detection result of the hatch sensor 48. When it is determined that the open-close state of the hatch 40 is changed, the process proceeds to Step S12.

(79) In Step S12, the ECU 39 determines whether the open-close state of the rack 50 is changed based on the detection result of the rack sensor 58. When it is determined that the open-close state of the rack 50 is changed, the process proceeds to Step S13.

(80) In Step S13, the ECU 39 controls the vehicle body height adjustment apparatus 36 to change the vehicle body height. The changed vehicle body height is preset based on the state of the hatch 40 or the state of the rack 50.

(81) For example, when it is determined in Step S11 and Step S12 that both the hatch 40 and the rack 50 are changed in the opening direction, the ECU 39 may control the vehicle body height adjustment apparatus 36 so that the vehicle body height gets lower in Step S13. For example, when it is determined in Step S11 and Step S12 that both the hatch 40 and the rack 50 are changed in a closing direction, the ECU 39 may control the vehicle body height adjustment apparatus 36 so that the vehicle body height gets higher in Step S13.

(82) FIG. 10 is a side view illustrating the vehicle 10 in which the vehicle body height is adjusted. FIG. 10 illustrates an example indicating an adjustment result of the vehicle body height in a case where it is determined that both the hatch 40 and the rack 50 are changed in the opening direction in Step S11 and Step S12.

(83) In the example illustrated in FIG. 10, the suspension 36 in the front part of the vehicle body 11 extends in a direction of an arrow mark A1, and the suspension 36 in the rear part of the vehicle body 11 shrinks in a direction of an arrow mark A2. Accordingly, the vehicle body height is

adjusted so that the front part is moved up and the rear part is moved down. The rear part of the vehicle body **11** where the rack **50** and the under-hatch component **32** are located is moved down, thus it is easy to have access to the stored object **60** mounted to the rack **50** or the under-hatch component **32**. The opening part **19** is also moved down, thus the under-hatch component **32** can be easily observed through the opening part **19**.

(84) For example, when it is determined in Step **S11** and Step **S12** that both the hatch **40** and the rack **50** are changed in the closing direction, the ECU **39** may change the vehicle body height from the state in FIG. **10** to the state in FIG. **1**.

(85) FIG. **11** is a flow chart illustrating another example of the control operation of the ECU **39**. In the example illustrated in FIG. **11**, the ECU **39** controls the vehicle body height adjustment apparatus **36** based on the state of the hatch **40**.

(86) In Step **S21**, the ECU **39** determines whether the open-close state of the hatch **40** is changed based on the detection result of the hatch sensor **48**. When it is determined that the open-close state of the hatch **40** is changed, the process proceeds to Step **S22**.

(87) In Step **S22**, the ECU **39** controls the vehicle body height adjustment apparatus **36** to change the vehicle body height. The change of the vehicle body height in Step **S22** may be any change described in Step **S13**. The control apparatus may lower the height of the vehicle body **11** upon determining that the hatch **40** is opened.

(88) FIG. **12** is a flow chart illustrating still another example of the control operation of the ECU **39**. In the example illustrated in FIG. **12**, the ECU **39** controls the vehicle body height adjustment apparatus **36** based on the state of the rack **50**.

(89) In Step **S31**, the ECU **39** determines whether the open-close state of the rack **50** is changed based on the detection result of the rack sensor **58**. When it is determined that the open-close state of the rack **50** is changed, the process proceeds to Step **S32**.

(90) In Step **S32**, the ECU **39** controls the vehicle body height adjustment apparatus **36** to change the vehicle body height. The change of the vehicle body height in Step **S32** may be any change described in Step **S13**. The control apparatus may lower the height of the vehicle body **11** upon determining that the rack **50** is inclined with respect to the vehicle body **11**.

(91) The control apparatus may adjust the vehicle body height based on the state of the vehicle **10** regarding the rack **50**. The state of the vehicle **10** regarding the rack **50** may include any one or some of the opening of the rack **50**, the opening of the hatch **40**, and the inclination of the rack **50**. Applicable as the adjustment of the vehicle body height is not only the case where the stored object **60** can be easily taken in and out but also a case where the posture of the rack **50** is set to get close to the horizontal posture so that the stored object **60** does not drop out even when the vehicle body **11** is inclined due to a concave-convex shape of a road surface.

(92) <Lock Control of Rack **50**>

(93) FIG. **13** is a block diagram illustrating an apparatus configuration of the vehicle **10**.

(94) The vehicle **10** includes the hatch sensor **48** described above, a rack lock **56**, and the control apparatus **39**. The hatch sensor **48** detects the rotation of the hatch **40** (release state of the hatch **40**). The rack lock **56** is configured to be able to switch lock and lock release of the rack **50**. For example, a position of the rack **50** where the lock and lock release are performed on the rack lock **56** may be the first position, the second position, and an intermediate position between the first position and the second position. The control apparatus **39** controls the rack lock **56**. Herein, the ECU **39** described above functions as the control apparatus **39** controlling the rack lock **56**.

(95) A detection result of the hatch sensor **48** is transmitted to the ECU **39**. The ECU **39** switches lock and lock release of the rack lock **56** based on the detection result of the hatch sensor **48**. For example, the ECU **39** releases the rack lock **56** to allow the rack **50** to move from the first position to the second position upon determining the release state of the hatch **40**.

(96) FIG. **14** is a plan view illustrating the lock/release state of the rack lock **56**.

(97) As illustrated in FIG. **14**, the rack lock **56** includes a latch **56A**, a striker **56B**, and a pivoting

part **56C**. One of the latch **56A** and the striker **56B** is provided to the rack body **51**, and the other one thereof is provided to a side of the vehicle body **11**. The other one thereof may be provided to the partition wall **22**, or may also be provided to the ROPS **12**, for example. The pivoting part **56C** pivots the latch **56A** in a direction of an arrow mark **A3**. Accordingly, the state of the latch **56A** is changed between a state of being locked to the striker **56B** and a state of being detached from the striker **56B**. The pivoting part **56C** is a motor, for example.

(98) FIG. **15** is a flow chart illustrating an example of a control operation of the ECU **39**.

(99) In Step **S41**, the ECU **39** determines whether the hatch **40** is opened based on the detection result of the hatch sensor **48**. When it is determined that the hatch **40** is opened, the process proceeds to Step **S42**.

(100) In Step **S42**, the ECU **39** controls the rack lock **56** and releases the rack lock **56**. Accordingly, the rack **50** can be moved from the first position to the second position.

(101) The vehicle **10** may include a load sensor detecting a load of the stored object **60** in the rack **50**. The ECU **39** may perform control so that the rack lock **56** is kept in the lock state when the load sensor detects the load of the stored object **60** in the rack **50**, and the lock of the rack lock **56** is released when the load sensor does not detect the load of the stored object **60** in the rack **50**.

Accordingly, dropping of the stored object **60** can be suppressed when the rack **50** is rotated.

(102) The vehicle **10** may include a sensor detecting an attachment state of the fixing component **52** in addition to the load sensor. In this case, when the load sensor detects the load of the stored object **60** in the rack **50** and also detects that the fixing component **52** is not attached, the ECU **39** may control so that the rack lock **56** is kept in the lock state. The ECU **39** may perform control so that the lock of the rack lock **56** is released when the load sensor does not detect the load of the stored object **60** in the rack **50**. The ECU **39** may perform control so that the lock of the rack lock **56** is released when the load sensor detects the load of the stored object **60** in the rack **50** and also detects that the fixing component **52** is attached. Accordingly, the rack **50** can be rotated more safely when the spare tire **60** is mounted to the tire rack **50** described above, for example.

(103) <Operating Element **55**>

(104) FIG. **16** is a side view illustrating a modification example of a position of the operating element **55**. As illustrated in FIG. **16**, the operating element **55** performing the open-close operation on the rack **50** may be located on an upper side of the rear wheels **30R**.

(105) In addition, the operating element **55** opening and closing the hatch **40** may also be provided. Both opening and closing of the hatch **40** and rotation of the rack **50** may be performed by operation of one operating element **55**.

(106) <Vehicle Body **11**>

(107) FIG. **17** is a side view illustrating a modification example of the vehicle body **11**.

(108) As illustrated in FIG. **17**, a vehicle body **111** may not include the hardtop **16**. In the example illustrated in FIG. **17**, the vehicle body **111** includes a hard cover **116** in place of the hardtop **16** described above. In the hard cover **116**, the roof part **17** is omitted in the hardtop **16** described above. In the hard cover **116**, an upper part of the front sidewall part **18A** is omitted in the hardtop **16** described above.

(109) The vehicle body may not also include the hard cover. The vehicle body may include a soft-top covering a region similar to the hardtop **16**. The vehicle body may include a soft cover covering a region similar to the hard cover. The vehicle body may not also include the cover covering the ROPS **12**.

(110) According to the vehicles **10** and **110** having the above configurations, the rack **50** is covered by the hatch **40**, thus the stored object **60** stored in the rack **50** can be protected from rain water, for example. In a state where the hatch **40** is closed and the rack **50** is located in the first position, the under-hatch component **32** can be protected by the rack **50** and the hatch **40**, and the stored object **60** loaded on the rack **50** can be protected by the hatch **40**. Furthermore, in the state where the hatch **40** is opened and the rack **50** is located in the second position, it is easy to have access to the

under-hatch component **32** through the opening part **19**.

(111) The under-hatch component **32** is the travel drive source **32**. The travel drive source **32** is a component requiring improvement of appearance or maintenance by exposure. The travel drive source **32** is exposed, thus maintenance properties of the travel drive source **32** are improved, and visibility of the design of the travel drive source **32** can be improved.

(112) The hatch **40** includes the hatch handle **43**. Accordingly, the open-close operation of the hatch body **41** can be easily performed by the handle.

(113) The rack hinge **53** is located in the rear end of the rack body **51**. Accordingly, the second position of the rack **50** is located outside the vehicle **10**.

(114) The hatch hinge **42** is located in a front end of the hatch body **41**, and the rack hinge **53** is located in the rear end of the rack body **51**. The hatch **40** and the rack **50** can be rotated in directions opposite to each other, thus hardly interfere with each other compared with a case where they are rotated in the same direction.

(115) The hatch **40** is moved to the upper position beyond the pivot range of the rack **50** while the opening part **19** is released. Thus, the interference between the hatch **40** and the rack **50** is prevented, and the rack **50** can be sufficiently inclined.

(116) The vehicle body **11** includes the partition wall **22** partitioning the luggage space in which the rack **50** is provided and the occupant space in which the seat **26** is provided. Accordingly, transfer of heat between the luggage space and the occupant space can be suppressed.

(117) The rack **50** is the tire rack **50** to which the spare tire **60** is mounted. The tire rack **50** can cover the under-hatch component **32**.

(118) The tire rack **50** can be moved between the first position and the second position while the spare tire **60** is mounted thereto. The spare tire **60** can be mounted to and taken out from the tire rack **50** in the second position, thus the spare tire **60** can be easily mounted and taken out.

(119) The vehicle **10** includes the vehicle body height adjustment apparatus **36** adjusting the height of the vehicle body **11**, and detects the rotation of the hatch **40**, thus the vehicle body height adjustment apparatus **36** adjusts the height of the vehicle body **11**. Accordingly, the vehicle body height can be automatically changed when the hatch **40** is rotated.

(120) The vehicle **10** includes the operating element **55** performing the open-close operation on the rack **50**, and the operating element **55** is located inside the vehicle **10** with respect to the hatch **40** in the close state. Suppressed accordingly is the operation of the operating element **55** while the hatch **40** is not opened.

(121) The operating element **55** performing the open-close operation on the rack **50** is located on the upper side of the rear wheels **30R** in the vehicle **110**. Accordingly, the operating element **55** can be operated even when the user does not approach the lateral side of the vehicle **10**.

(122) The vehicle **10** includes the hatch holder **46** holding the hatch **40** in the position where the hatch **40** is opened. Accordingly, the hatch **40** can be held in the open state.

(123) The vehicle **10** includes the rack holder **54** holding the rack **50** in the second position. Accordingly, the rack **50** can be held in a state of being located in the second position.

(124) The hatch **40** is supported by the ROPS **12**. Accordingly, the hatch **40** is easily protected by the ROPS **12**.

(125) The hatch hinge **42** is connected to the ROPS **12**. Accordingly, the hatch **40** is easily supported by the ROPS **12**.

(126) The rack hinge **53** is connected to the ROPS **12**. Accordingly, the rack **50** is easily supported by the ROPS **12**.

(127) The vehicle **10** includes the hatch sensor **48** determining the release state of the hatch **40** and the rack lock **56** allowing the rack **50** to move from the first position to the second position when the sensor **48** determines the release state of the hatch **40**. Accordingly, movement of the rack **50** from the first position to the second position can be suppressed while the hatch **40** is in the close state.

(128) The control apparatus may control the vehicle body height adjustment apparatus 36 based on the state of the vehicle 10 regarding the rack 50 to adjust the height of the vehicle body 11. In this case, the vehicle body height adjustment apparatus 36 is controlled based on the state of the vehicle 10 regarding the rack 50 and the height of the vehicle body 11 is adjusted, thus the stored object 60 can be easily taken in and out from the rack 50, and can be stored in an appropriate state in the rack 50.

(129) The control apparatus 39 may lower the height of the vehicle body 11 upon determining that the rack 50 is inclined with respect to the vehicle body 11. Accordingly, movement of the rack 50 from the first position to the second position can be suppressed while the hatch 40 is in the close state.

(130) This application discloses the following aspects.

(131) A first aspect is a vehicle including: a vehicle body including an opening part located in a rear part of the vehicle; a hatch supported by the vehicle body so as to be able to open and close the opening part; an under hatch component located inside the vehicle with: respect to the opening part; and a rack movably supported by the vehicle body between a first position facing the under-hatch component and a second position retracted from the first position, wherein the under-hatch component is exposed outside through the opening part in a state where the hatch is opened and the rack is located in the second position.

(132) According to the present vehicle, the rack is covered by the hatch, thus the stored object stored in the rack can be protected from rain water, for example. In the state where the hatch is closed and the rack is located in the first position, the under-hatch component can be protected by the rack and the hatch, and the stored object loaded on the rack can be protected by the hatch. Furthermore, in the state where the hatch is opened and the rack is located in the second position, it is easy to have access to the under-hatch component through the opening part.

(133) A second aspect is the vehicle according to the first aspect, wherein the under-hatch component is a travel drive source.

(134) In this case, the travel drive source is the component requiring improvement of appearance or maintenance by exposure. The travel drive source is exposed, thus maintenance properties of the travel drive source are improved, and visibility of the design of the travel drive source can be improved.

(135) A third aspect is the vehicle according to the first or second aspect, wherein the hatch includes a hatch body, a hatch hinge located in a front end of the hatch body to be connected to the vehicle body, and a hatch handle located in a rear part of the hatch body.

(136) Accordingly, in this case, the open-close operation of the hatch body can be easily performed by the handle.

(137) A fourth aspect is the vehicle according to any one of the first to third aspects, wherein the rack includes a rack body and a rack hinge located in a rear end of the rack body to be connected to the vehicle body.

(138) In this case, the second position of the rack is located outside the vehicle.

(139) A fifth aspect is the vehicle according to any one of the first to fourth aspects, wherein the hatch includes a hatch body and a hatch hinge located in a front end of the hatch body to be connected to the vehicle body, and the rack includes a rack body and a rack hinge located in a rear end of the rack body to be connected to the vehicle body.

(140) In this case, the hatch and the rack can be rotated in directions opposite to each other, thus hardly interfere with each other compared with a case where they are rotated in the same direction.

(141) A sixth aspect is the vehicle according to the fifth aspect, wherein the hatch is moved to an upper position beyond a pivot range of the rack while the opening part is released.

(142) Accordingly, interference between the hatch and the rack is prevented, and the rack can be sufficiently inclined.

(143) A seventh aspect is the vehicle according to any one of the first to sixth aspects, wherein the

vehicle body includes a partition wall partitioning a luggage space in which the rack is provided and an occupant space in which a seat is provided.

(144) Accordingly, transfer of heat between the luggage space and the occupant space can be suppressed.

(145) An eighth aspect is the vehicle according to any one of the first to seventh aspects, wherein the rack is a tire rack to which a spare tire is mounted.

(146) In this case, the tire rack can cover the under-hatch component.

(147) A ninth aspect is the vehicle according to the eighth aspect, wherein the tire rack can be moved between the first position and the second position while the spare tire is mounted.

(148) In this case, the spare tire can be mounted to and taken out from the tire rack in the second position, thus the spare tire can be easily mounted and taken out.

(149) A tenth aspect is the vehicle according to any one of the first to ninth aspects, further including a vehicle body height adjustment apparatus adjusting a height of the vehicle body, wherein the vehicle body height adjustment apparatus adjusts the height of the vehicle body upon detecting rotation of the hatch.

(150) Accordingly, the vehicle body height can be automatically changed when the hatch is rotated.

(151) An eleventh aspect is the vehicle according to any one of the first to tenth aspects, further including an operating element located inside the vehicle with respect to the hatch in a close state and performing an open-close operation on the rack.

(152) The operation of the operating element can be suppressed while the hatch is not opened.

(153) A twelfth aspect is the vehicle according to any one of the first to eleventh aspects, further including: a front wheel and a rear wheel; and an operating element located on an upper side of the rear wheel and performing an open-close operation on the rack.

(154) Accordingly, the operating element can be operated even when a user does not approach a lateral side of the vehicle.

(155) A thirteenth aspect is the vehicle according to any one of the first to twelfth aspects, further including a hatch holder holding the hatch in a position where the hatch is opened.

(156) Accordingly, the hatch can be held in the open state.

(157) A fourteenth aspect is the vehicle according to any one of the first to thirteenth aspects, further including a rack holder holding the rack in the second position.

(158) Accordingly, the rack can be held in a state of being located in the second position.

(159) A fifteenth aspect is the vehicle according to any one of the first to fourteenth aspects, wherein the hatch is supported by a rollover protective structure (ROPS) determining an occupant space.

(160) Accordingly, the hatch is easily protected by the ROPS.

(161) A sixteenth aspect is the vehicle according to the fifteenth aspect, wherein the hatch includes a hatch body and a hatch hinge located in an end portion of the hatch body to be connected to the ROPS.

(162) Accordingly, the hatch is easily supported by the ROPS.

(163) A seventeenth aspect is the vehicle according to the fifteenth or sixteenth aspect, wherein the rack includes a rack body and a rack hinge located in an end portion of the rack body to be connected to the ROPS.

(164) Accordingly, the rack can be easily supported by the ROPS.

(165) An eighteenth aspect is the vehicle according to any one of the first to seventeenth aspects, further including: a sensor determining a release state of the hatch; and a lock mechanism of allowing the rack to move from the first position to the second position when the sensor determines the release state of the hatch.

(166) Accordingly, movement of the rack from the first position to the second position can be suppressed while the hatch is in the close state.

(167) A nineteenth aspect is a vehicle including: a vehicle body; a vehicle body height adjustment

apparatus adjusting a height of the vehicle body; a rack provided in a rear part of the vehicle body and provided with a stored object; and a control apparatus controlling the vehicle body height adjustment apparatus based on a vehicle state regarding the rack and adjusting the height of the vehicle body.

(168) According to the present vehicle, the vehicle body height adjustment apparatus is controlled based on the vehicle state regarding the rack and the height of the vehicle body is adjusted, thus the stored object can be easily taken in and out from the rack, and can be stored in an appropriate state in the rack.

(169) A twentieth aspect is the vehicle according to the nineteenth aspect, wherein the control apparatus lowers the height of the vehicle body upon determining an inclination of the rack with respect to the vehicle body.

(170) Accordingly, movement of the rack from the first position to the second position can be suppressed while the hatch is in the close state.

(171) The foregoing description is in all aspects illustrative and does not restrict the present invention. Numerous modifications and variations that have not yet been exemplified are devised without departing from the scope of the present invention.

Claims

1. A vehicle, comprising: a vehicle body including an opening part located in a rear part of the vehicle; a hatch supported by the vehicle body so as to be able to open and close the opening part; an under hatch component located inside the vehicle with respect to the opening part; and a rack movably supported by the vehicle body between a first position facing the under-hatch component and a second position retracted from the first position, wherein the under-hatch component is a travel drive source, the rack is to be a partition wall covering an upper side of the travel drive source in a state of being in the first position, and the travel drive source is exposed outside through the opening part in a state where the hatch is opened and the rack is located in the second position.
2. The vehicle according to claim 1, wherein the hatch includes a hatch body, a hatch hinge located in a front end of the hatch body to be connected to the vehicle body, and a hatch handle located in a rear part of the hatch body.
3. The vehicle according to claim 1, wherein the rack includes a rack body and a rack hinge located in a rear end of the rack body to be connected to the vehicle body.
4. The vehicle according to claim 1, wherein the hatch includes a hatch body and a hatch hinge located in a front end of the hatch body to be connected to the vehicle body, the rack includes a rack body and a rack hinge located in a rear end of the rack body to be connected to the vehicle body, and the rack hinge is behind the hatch hinge.
5. The vehicle according to claim 4, wherein the hatch is moved to an upper position beyond a pivot range of a front end of the rack while the opening part is released.
6. The vehicle according to claim 1, wherein the vehicle body includes a partition wall partitioning a luggage space in which the rack is provided and an occupant space in which a seat is provided.
7. The vehicle according to claim 1, wherein the rack is a tire rack to which a spare tire is mounted, and the tire rack can be moved between the first position and the second position while the spare tire is mounted.
8. The vehicle according to claim 1, further comprising a vehicle body height adjustment apparatus adjusting a height of the vehicle body, wherein the vehicle body height adjustment apparatus adjusts the height of the vehicle body upon detecting rotation of the hatch.
9. The vehicle according to claim 1, further comprising an operating element located inside the vehicle with respect to the hatch in a close state and performing an open-close operation on the rack.
10. The vehicle according to claim 1, further comprising: a front wheel and a rear wheel; and an

operating element located on an upper side of the rear wheel and performing an open-close operation on the rack.

11. The vehicle according to claim 1, further comprising a hatch holder holding the hatch in a position where the hatch is opened.

12. The vehicle according to claim 1, wherein the rack includes a rack body and a rack hinge located in a rear end of the rack body to be connected to the vehicle body, the vehicle further comprising a rack holder holding the rack in the second position, wherein the rack holder includes a piston connected to the rack and the vehicle body.

13. The vehicle according to claim 1, wherein the hatch is supported by a rollover protective structure (ROPS) determining an occupant space.

14. The vehicle according to claim 13, wherein the hatch includes a hatch body and a hatch hinge located in an end portion of the hatch body to be connected to the ROPS.

15. The vehicle according to claim 13, wherein the rack includes a rack body and a rack hinge located in an end portion of the rack body to be connected to the ROPS.

16. The vehicle according to claim 1, further comprising: a sensor determining a release state of the hatch; and a lock mechanism of allowing the rack to move from the first position to the second position when the sensor determines the release state of the hatch.
